TIGER ANALYTICS APP EGINEERING ASSIGNMENT – Design Decision DOCUMENT

<u>Note: While the format of design decision document varies project to project and it depends on the expectations and I have much experience in preparing Design Document based on Java classes than the design decision kind of documents. Hence I am consolidating the plans I had while creating this module as design considerations.</u>

1. Combined Architecture (MySQL + Elasticsearch) — MySQL serves as the DB for all writes, while Elasticsearch provides fast & efficient full-text search. If Elasticsearch goes down, data is safe in MySQL and the index can be rebuilt.
2. UUID Primary Keys — UUIDs enable ID generation without database coordination and prevent record enumeration attacks since IDs are non-sequential.
3. Upsert on Duplicate Upload — When a CSV contains a record matching an existing (store_id, sku, date) combination, the system updates the existing record instead of rejecting or duplicating it. This allows stores to upload corrected feeds without manual cleanup.
4. Partial CSV Processing — Valid rows are saved even if other rows have errors. A detailed per-row error report is returned so operators can fix only the failed rows instead of re-uploading the entire file.
5. Elasticsearch Custom Analyzer with Asciifolding — A custom product_name_analyzer with lowercase and asciifolding filters enables accent-insensitive search (e.g., "cafe" matches "Cafe Latte Powder"), which is essential for a multi-country retail chain.
6. Separate Repository Packages for JPA and ES — JPA and Elasticsearch repositories are placed in distinct packages with explicit @Enable annotations to prevent Spring Data module detection conflicts at startup.
7. Angular 21 Standalone Components — Each component is self-contained with its own imports, eliminating the need for a central NgModule. This is the standard pattern for modern Angular and enables better tree-shaking.
8. Server-Side CORS Configuration — CORS is configured in the Spring Boot backend (not via Angular proxy), making the access policy explicit and portable across environments.

Prepared by: Jeyasree S
Prepared on: 09-Feb-2026